Mathmatical Tools and Their Use In Physics

Building Models to Describe our World

Vectors and Calculus

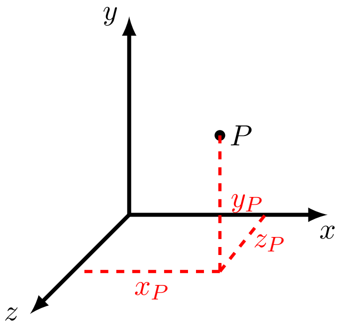
Outline

Vectors

Mathmatical Tools and Their Use In Physics

Coordinate systems

- Because we want to describe things in space, and we live in 3 dimensions!
- In practice, we can often simplify and use 1D or 2D coordinate systems.
- The most common type of coordinate system is a right-handed Cartesian coordinate system, where you must specify:
  - 3 mutually perpendicular axes/directions (specify 2, the third is given by the right hand rule).
  - 1 fixed origin.
- Using a coordinate system, can specify the position of a point using its **coordinates**.
- Can you think of an example where you've used spherical coordinates?

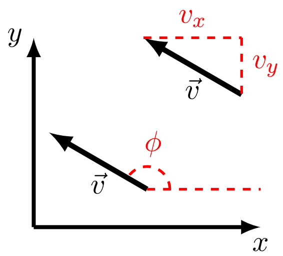


A ***right-handed*** *Cartesian coordinate system, where $\hat{x} \times \hat{y} = \hat{z}$. ($\hat{z}$ is out of the page). The point P has coordinates (x_P, y_P, z_P).*

Vectors

- Vectors are arrows with a magnitude and a direction.
- They are useful for describing quantities with a magnitude and a direction:
  - A displacement.
  - A velocity.
  - A force.
  - An angular velocity.
- We can describe a vector by specifying its **components**, or its magnitude and direction(s) (in 3D, need magnitude and two angles).
- Notation:

$$\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z} = (v_x, v_y, v_z) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$$



A vector in 2D can be specified by two Cartesian components (v_x, v_y) or its length, v, and an angle with respect to the x direction, ϕ.

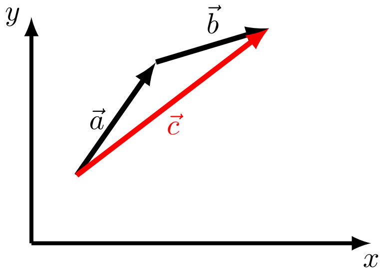
Algebra with vectors

- You can multiply/divide a vector by **a scalar**.
- You cannot add a scalar to a vector (as with anything, you cannot add oranges and apples).
- You can add/subtract two vectors (of the same dimensions, e.g. 2D or 3D).
- Two ways to multiply vectors:
  - Scalar/dot product $\rightarrow$ the result is a scalar.
  - Vector/cross product $\rightarrow$ the results is a vector (and this requires 3D and a flexible wrist).

Vector addition and vector equations

- To add vectors $\vec{a}$ and $\vec{b}$:
  - Graphically, place them head to tail.
  - Algebraically, add the components together.
- The "unit vector" notation for a vector is just vector addition:

$$\vec{a} = a_x\hat{x} + a_y\hat{y}$$



A graphical illustration of $\vec{c} = \vec{a} + \vec{b}$.

Vector addition and vector equations

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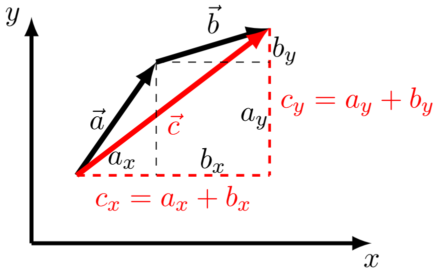
- **An equation with vectors is a short-hand for equations between components:**

$$\vec{c} = \vec{a} + 2\vec{b}$$

is the same as the two independent equations:

$$c_x = a_x + 2b_x$$
$$c_y = a_y + 2b_y$$



A graphical illustration of $\vec{c} = \vec{a} + \vec{b}$.

Scalar product

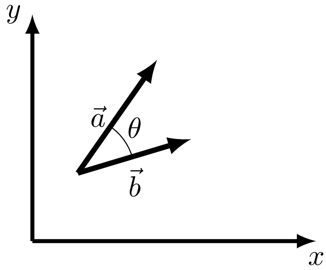
- Scalar product is a sum of the components of the two vectors multiplied together (commutative):

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z c_z = \vec{b} \cdot \vec{a}$$

- It is also proportional to the cosine of the angle between the vectors (if these are placed tail to tail):

$$\vec{a} \cdot \vec{b} = ab \cos \theta$$

- It is thus zero if two vectors are perpendicular to each other, $\cos(90) = 0$
- By knowing the components of two vectors, you can determine their scalar product, and then the angle between the two vectors



The scalar product between vectors depends on the angle between them when they are placed tail to tail.

Cross-product

- The cross product gives a **vector** that is perpendicular to the original two vectors (not commutative):

$$\vec{a} \times \vec{b} = \vec{c} = -\vec{b} \times \vec{a}$$

- The magnitude is related to the sine of the angle between the two original vectors:

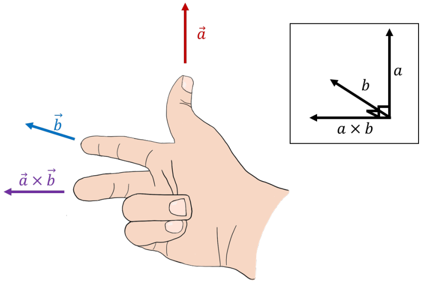
$$||\vec{a} \times \vec{b}|| = ab \sin\theta$$

- The cross-product is zero if the two vectors are parallel or anti-parallel ($\sin 0 = 0$)
- The direction of the cross product can be found using the right hand rule
- Cross products are always in 3D

You can find the components of the cross product algebraically:

$$\vec{a} \times \vec{b} = \begin{pmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{pmatrix}$$

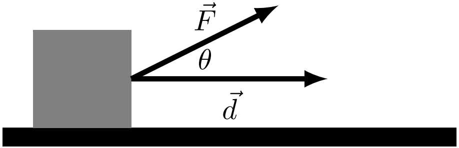
or using the right-hand rule for cross-products for the direction:



Scalar and cross product in action

Scalar product:

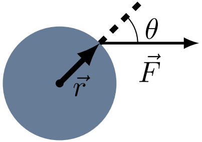
- Maximal when two vectors are parallel.
- Zero when two vectors are perpendicular.
- Application: “Work”.



The scalar product between force and displacement, $W = \vec{F} \cdot \vec{d} = F d \cos\theta$ *(Work) tells us about the change in speed.*

Cross product:

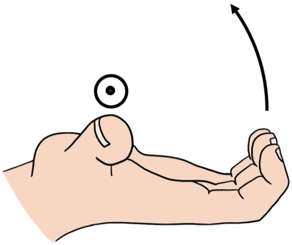
- Maximal when two vectors are perpendicular.
- Zero when two vectors are parallel.
- Application: “Torque”.

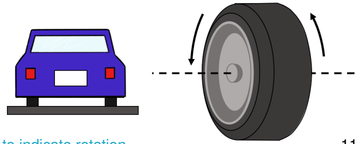


The magnitude of cross product between a force, $\vec{F}$*, and the position vector,* $\vec{r}$*, where the force is applied tells us the angular acceleration of an object:* $||\vec{r} \times \vec{F}|| = rF\sin\theta$.

Vectors to indicate rotation

- Suppose that you need to describe a rotating wheel to someone over the phone.
- You need to specify:
  - The axis of rotation (a line)
  - How fast (e.g. rotations per minute)
  - Which direction (clockwise/counter clockwise) about that axis
- This can be specified by an “axial vector”, and the “right hand rule for axial vectors”:
  - The vector is co-linear with the axis of rotation
  - The magnitude is proportional to rotational speed
  - The direction of the vector is given by the right hand rule for axial vectors.





Derivatives

Mathmatical Tools and Their Use In Physics

Functions

- We will mostly concern ourselves with functions that are mappings from a set of independent variables (x, y , z, t) to (usually) a real number.
- Examples of functions in physics:
  - Position as a function of time for an accelerating object: $x(t) = x_0 + v_{x0}t + \frac{1}{2}a_x t^2$
  - The restoring force from a spring: $F(x) = -kx$
  - The potential energy from gravity near Earth's surface: $V(h) = mgh$
  - The electric field from a point charge at the origin:

$$\vec{E}(x, y, z) = \frac{kq}{x^2 + y^2 + z^2}\hat{r}$$ (this is one is not a mapping to a real number!)

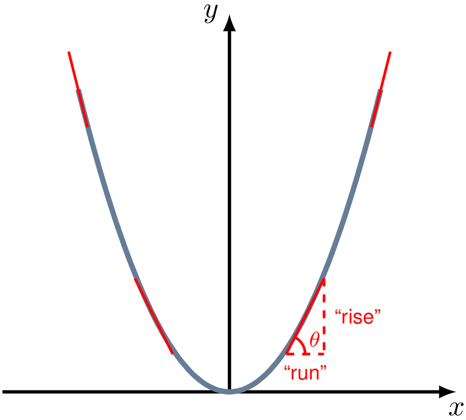
  - etc. . . .

Properties of a function

- When you encounter a function, ask yourself:
  - Is the function ever equal to zero (what are its “roots”)?
  - Is the function continuous for all values of the independent variables?
  - Are there maxima/minima in the function?
  - Is the function always defined? This is especially important in physics. We need functions to represent physical quantities, so we need to understand when they are undefined
- The best thing to do is to sketch out the function to get a sense of it $\rightarrow$ There are many online tools (e.g. Wolfram alpha, Jupyter notebooks, etc)

The derivative

- The derivative of a function, $f(x)$, is the slope of the function (“rise over run”).
- It tells us how steep the function is, namely, how fast it is increasing or decreasing, or neither $\rightarrow$ the rate of change of $f(x)$ with respect to x.
- The derivative of a function is a function.
- Different notations for the derivative of $f(x)$ with respect to x:
  $f'(x)$, $\frac{df}{dx}$, $\frac{d}{dx}f(x)$
- It tells us the (tangent of the) angle which the tangent to the function makes with the x-axis.

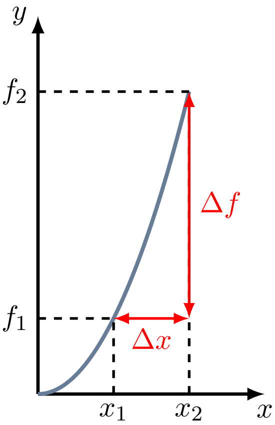


The derivative of the function is given by the “rise over the run”. This corresponds to the tangent of the angle that the tangent to the function makes at that point.

The derivative is the *instantaneous* rate of change of a function

- It tells us how much $f(x)$ changes, when we change x, at some point x.
- That is, it tells us the value of Δf if we go from x to $x + \Delta x$.
- Suppose that we choose two points, x_1 and x_2, separated by a distance Δx:

$$\Delta x = x_2 - x_1, \quad f(x_1) = f_1, \quad f(x_2) = f_2$$
$$\Delta f = f_2 - f_1$$
$$\therefore f'(x) = \frac{df}{dx} = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$



The derivative of the function is given by $\frac{\Delta f}{\Delta x}$ in the limit where $\Delta x \to 0$.

How to find a derivative

- You'll learn in calculus (the textbook shows how to do derive it for the case of $f(x) = x^2$).
- For now, you need to know where to look them up, and how to apply the rules:

Function, $f(x)$	**Derivative**, $f'(x)$
$f(x) = a$	$f'(x) = 0$
$f(x) = x^n$	$f'(x) = nx^{n-1}$
$f(x) = \sin(x)$	$f'(x) = \cos(x)$
$f(x) = \cos(x)$	$f'(x) = -\sin(x)$
$f(x) = \tan(x)$	$f'(x) = \frac{1}{\cos^2(x)}$
$f(x) = e^x$	$f'(x) = e^x$
$f(x) = \ln(x)$	$f'(x) = \frac{1}{x}$

Common derivatives of functions.

How to find a derivative

- You'll learn in calculus (the textbook shows how to do derive it for the case of $f(x) = x^2$).
- For now, you need to know where to look them up, and how to apply the rules:

Function, $h(x)$	**Derivative**, $h'(x)$
$h(x) = f(x) + g(x)$	$h'(x) = f'(x) + g'(x)$
$h(x) = f(x) - g(x)$	$h'(x) = f'(x) - g'(x)$
$h(x) = f(x)g(x)$	$h'(x) = f'(x)g(x) + f(x)g'(x)$ (The product rule)
$h(x) = \frac{f(x)}{g(x)}$	$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$ (The quotient rule)
$h(x) = f(g(x))$	$h'(x) = f'(g(x))g'(x)$ (The Chain Rule)

Derivatives of combined functions.

Inifinitesimals - Calculus notation

- Calculus usually involves asking what happens when something is very small
- The derivative is:

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x}$$

- We write this as:

$$f'(x) = \frac{df}{dx}$$

- You can really think of dx as a very small Δx, and df as the *correspondingly* small Δf.
- You can multiply them out:

$$f'(x)dx = df$$

- You can cancel them out:

$$\frac{df}{dg}\frac{dg}{dx} = \frac{df}{dx}$$

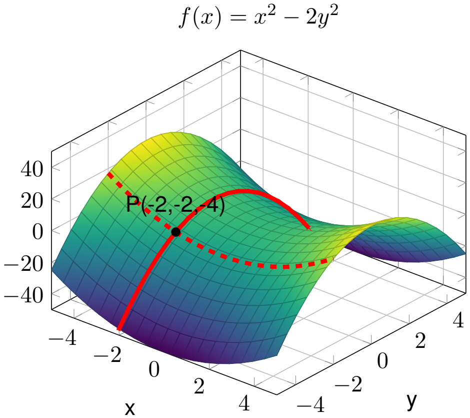
- This works because

$$dx = \lim_{\Delta x \to 0} \Delta x$$

(your math friends are cringing. . .)

Functions of multiple variables

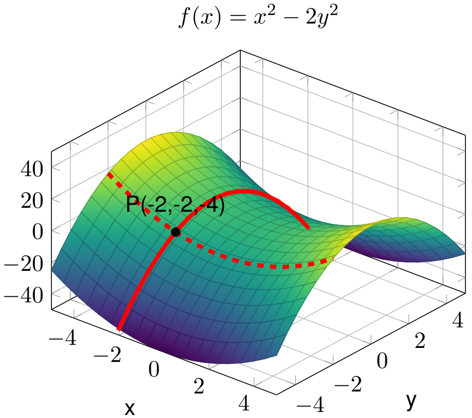
- Many quantities depend on where you are in space (i.e. a function of x,y,z).
- These become difficult to visualize...
- We have to define the rate of change of the function in any direction...
- In the plot, at point P:
    - The function decreases as x increases (if y is held fixed, dashed red line).
    - The function increases as y increases (if x is held fixed, solid red line).



A point, P, is on the surface given by $f(x, y) = x^2 - 2y^2$.

Functions of multiple variables

- The partial derivative of $f(x, y)$ with respect to x tells us the rate of change of f as a function of x if we hold y constant (following the dashed line).
- The partial derivative of $f(x, y)$ with respect to y tells us the rate of change of f as a function of y if we hold x constant (following the solid line).
- There are n partial derivatives if there are n independent variables.
- Each partial derivative is a new function of the n variables.



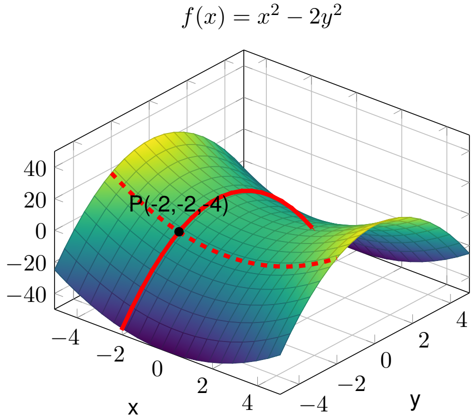
A point, P, is on the surface given by $f(x, y) = x^2 - 2y^2$.

Functions of multiple variables

Notation:

$$f(x,y) = x^2 - 2y^2$$
$$\therefore \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} x^2 - 2y^2 = 2x$$
$$\therefore \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} x^2 - 2y^2 = -4y;$$

- Pronounced “die f by die x”.
- Calculate a partial derivative exactly the same way that you would a normal derivative (treat all other independent variables as constants).

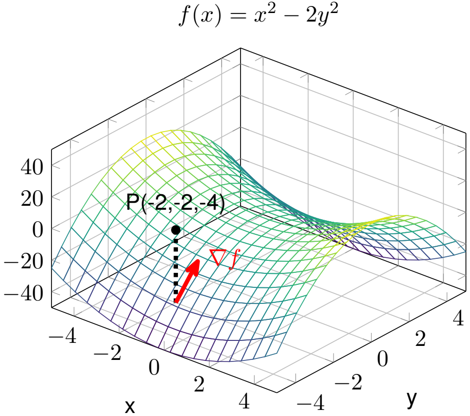


A point, P, is on the surface given by $f(x,y) = x^2 - 2y^2$.

The gradient vector

- From the partial derivatives, we know how the function changes along the x and y directions.
- The gradient tells us the direction in which the function changes the fastest, with components given by the partial derivatives.
- It is a vector whose magnitude tells us how fast the function is changing in the direction of maximal change.

$$\nabla f(x, y) = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y}$$
$$= 2x\hat{x} - 4y\hat{y} = (-4, 8)$$



A vector parallel to the gradient vector is projected in the xy-plane. It points in the direction of most rapid increase.

Integrals

Mathmatical Tools and Their Use In Physics

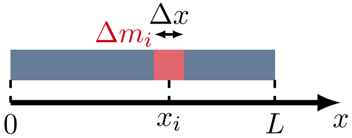
Example: The mass of a uniform rod

- Suppose that the rod has length L and mass M
- We can define a linear mass density, μ, (mass per unit length):

$$\mu = \frac{M}{L}$$

- A small section of rod, with length, Δx , will have a mass:

$$\Delta m_i = \mu \Delta x$$



A rod of mass, M, and length, L, modeled as the sum of many "mini rods" of length Δx and mass Δm_i.

Example: The mass of a uniform rod

- Suppose that the rod has length L and mass M
- We can define a linear mass density, μ, (mass per unit length):

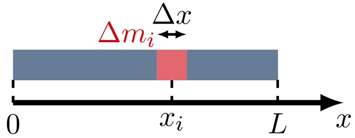
$$\mu = \frac{M}{L}$$

- A small section of rod, with length,Δx , will have a mass:

$$\Delta m_i = \mu \Delta x$$

- We can think of the total rod of mass M, as the sum of N identical "mini rods" of length, Δx , each with mass, Δm :

$$M = \sum_{i=0}^{i=N-1} \Delta m_i = \sum_{i=0}^{i=N-1} \mu \Delta x = \mu \sum_{i=0}^{i=N-1} \Delta x = \mu L$$



A rod of mass, M, and length, L, modeled as the sum of many "mini rods" of length Δx and mass Δm_i.

Example: The mass of a *non-uniform* rod

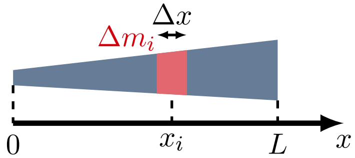
- Suppose the linear density *varies* with position:

$$\mu = \mu(x)$$

- To get the mass of a length, L, of rod, we have to sum the masses of N *different* mini-rods:

$$\Delta m_i = \mu(x_i)\Delta x$$
$$\therefore M = \sum_{i=0}^{i=N-1} \Delta m_i = \lim_{\Delta x \to 0} \sum_{i=0}^{i=N-1} \mu(x_i)\Delta x$$

- How do we evaluate this sum? As $\Delta x \to 0$, there are an infinite number of mini-rods (an infinite number of terms in the sum).
- Keep in mind that this is the problem that we'd like to solve, as we go and get lost in math.



A non-uniform rod of mass, M, and length, L, modeled as the sum of many "mini rods" of length Δx and mass Δm_i.

The anti-derivative

- Tuesday, we saw that we can determine the rate of change of a function with respect to one (or more) of its independent variables (the derivative).
- Today, we will see that given the derivative of a function, we can find the original function (the anti-derivative of the function that we are given).
- Derivative of $f(x)$ with respect to x:

$$f'(x) = \frac{d}{dx} f(x) \quad \text{Take the derivative with respect to x.}$$

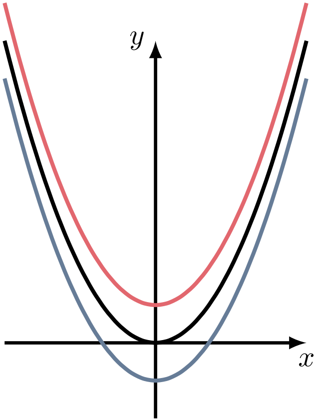
- Anti-derivative of $f(x)$ with respect to x:

$$F(x) = \int f(x) dx \quad \text{Take the anti-derivative with respect to x.}$$

$$\therefore f(x) = \frac{d}{dx} F(x)$$

The constant, C

- Anti-derivatives are only determined up to a constant, C.
- A synonym of anti-derivative is "indefinite integral" (since the function is not quite defined unless we specify C).
- In physics, we need functions that describe physical quantities (say, your position along the x-axis as a function of time); those functions need to be fully defined (not up to some constant C).
- The choice of the constant C effectively chooses a point through which the function must pass.
- It makes sense that the anti-derivative is not fully determined, since all of the functions on the right have the same derivative.



These functions, $f(x) = x^2 + C$ all have the same derivative but different values of $C = \{-0.5, 0, 0.5\}$.

Determining the anti-derivative of a function

- We would like to determine a function, $F(x)$, knowing:
  - its derivative, $f(x)$.
  - the function must go through a specific point, say (x_0, F_0).
- Physics analogy: we would like to know the position of an object as a function of time, $x(t)$, knowing
  - the velocity of the object as a function of time, $v(t)$ (the time derivative of the position):

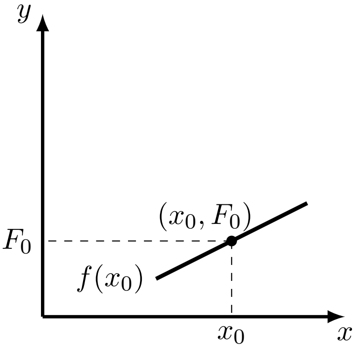
$$v(t) = \frac{d}{dt}x(t)$$

  - at time $t = t_0$, the object was located at a specific position:

$$x(t_0) = x_0$$

Graphically, how to visualize an anti-derivative

- We know:
  - the function $F(x)$ must go through a specific point, (x_0, F_0) .
  - the derivative (slope) of the function at that point, $f(x_0)$.



The "shooting method" for the anti-derivative.

Graphically, how to visualize an anti-derivative

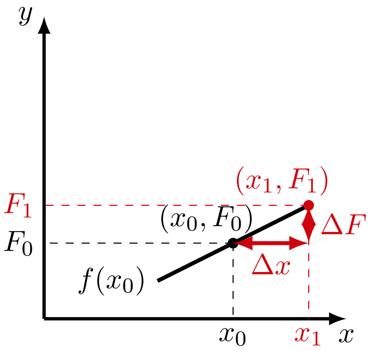
- We know:
  - the function $F(x)$ must go through a specific point, (x_0, F_0) .
  - the derivative (slope) of the function at that point, $f(x_0)$.
- We can determine the next point, (x_1, F_1), through which the function should pass:

$$\frac{\Delta F}{\Delta x} \approx f(x_0) \quad \text{becomes equal in the} \lim_{\Delta x \to 0}$$

$$\therefore \Delta F \approx f(x_0)\Delta x$$

$$\therefore F_1 \approx F_0 + \Delta F = F(x_0) + f(x_0)\Delta x$$

→ We now have a method to iterate and find all the points of $F(x)$.



The "shooting method" for the anti-derivative.

Extending the sum

- We have a procedure that allows us to determine the value of the function, $F(x)$, at any value of x (say x_N):

$$F(x_N) = F(x_0) + f(x_0)\Delta x + f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_{N-1})\Delta x$$

  where $x_N = x_0 + N\Delta x$

- Using the summation notation and taking the limit:

$$F(x_N) = F(x_0) + \lim_{\Delta x \to 0} \sum_{i=0}^{N-1} f(x_i)\Delta x$$

The indefinite and definite integrals

- We can re-write the sum as the difference in the anti-derivative evaluated at x_N and x_0:

$$F(x_N) - F(x_0) = \lim_{\Delta x \to 0} \sum_{i=0}^{N-1} f(x_i)\Delta x$$

  The constant, C, in the anti-derivative cancels when we take this difference.

- We call this difference the "**definite integral (or just integral) of $f(x)$ from x_0 to x_N**", and use the following notation:

$$\boxed{F(x_N) - F(x_0) = \int_{x_0}^{x_N} f(x)dx}$$

- Note the "limits" on the integral, which differentiates this from the anti-derivative.

Definite vs indefinite

- Given the function, $f(x) = x^2$
- The indefinite integral (or anti-derivative) is a **function**:

$$F(x) = \int f(x)dx = \int x^2 dx = \frac{1}{3}x^3 + C$$

- The definite integral between $x = 1$ and $x = 3$ is a **number**:

$$\int_1^3 x^2 dx = F(3) - F(1) = \frac{1}{3}3^3 - \frac{1}{3}1^3 = 9 - \frac{1}{3} = \frac{26}{3}$$

  Note that the C cancels in the subtraction.
- Physical quantities in physics must be given by definite integrals.

How to determine the anti-derivative

- To determine the anti-derivative of a function, $f(x)$, we need to be able to evaluate the sum:

$$\lim_{\Delta x \to 0} \sum_{i=0}^{N-1} f(x_i)\Delta x$$

- The textbook shows you how to do it for $f(x) = 2x$. Your calculus course will show you how to do it for other functions...

Function, $f(x)$	**Anti-derivative,** $F(x)$
$f(x) = a$	$F(x) = ax + C$
$f(x) = x^n$	$F(x) = \frac{1}{n+1}x^{n+1} + C$
$f(x) = \frac{1}{x}$	$F(x) = \ln(\|x\|) + C$
$f(x) = \sin(x)$	$F(x) = -\cos(x) + C$
$f(x) = \cos(x)$	$F(x) = \sin(x) + C$
$f(x) = \tan(x)$	$F(x) = -\ln(\|\cos(x)\|) + C$
$f(x) = e^x$	$F(x) = e^x + C$
$f(x) = \ln(x)$	$F(x) = x\ln(x) - x + C$

Common indefinite integrals of functions. Unlike derivatives, for which you can always determine an analytical function, you cannot always determine an analytical form for the indefinite integral of a function.

Graphical representation of the sum

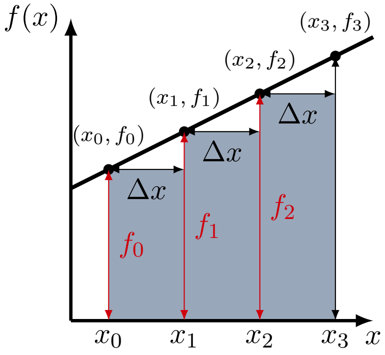
- If we graph $f(x)$ we can see that each term in the sum for the integral corresponds to the area of a little rectangle under the function $f(x)$:

$$\lim_{\Delta x \to 0} \sum_{i=0}^{N-1} f(x_i)\Delta x = f(x_0)\Delta x + f(x_1)\Delta x$$
$$+ f(x_2)\Delta x + \cdots + f(x_{N-1})\Delta x$$

- The definite integral:

$$\int_{x_0}^{X_N} f(x)dx = \lim_{\Delta x \to 0} \sum_{i=0}^{N-1} f(x_i)\Delta x$$

is thus the total area underneath $f(x)$, between x_0 and x_N



Each rectangle has an area $f(x_i)\Delta x$. As $\Delta x \to 0$, the rectangles correspond to the area under $f(x)$.

Back to physics: The mass of a *non-uniform* rod

- Suppose the linear density *varies* with position:

$$\mu = \mu(x)$$

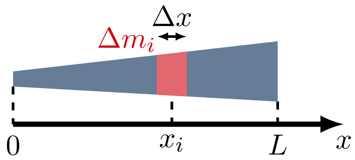
- To get the mass of a length L of rod, we have to sum the masses of N *different* mini-rods:

$$M = \sum_{i=0}^{i=N-1} \Delta m_i = \lim_{\Delta x \to 0} \sum_{i=0}^{i=N-1} \mu(x_i)\Delta x$$

- The sum, is given by the definite integral:

$$\lim_{\Delta x \to 0} \sum_{i=0}^{i=N-1} \mu(x_i)\Delta x = \int_{x=0}^{x=L} \mu(x)dx$$

If we know $\mu(x)$, we can ask our math friends for the anti-derivative of that function.



A non-uniform rod of mass, M, and length, L, modeled as the sum of many "mini rods" of length Δx and mass Δm_i.

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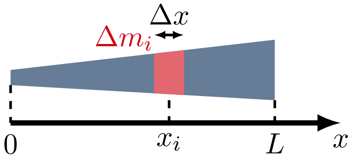

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As physicists, we need a tool to be able to calculate these sums. The mathematicians (well actually...) have developed this tool from calculus that can find the anti-derivative of a function. It turns out that this tool also gives a way to calculate the sums that we need in physics!

Back to physics: The mass of a *non-uniform* rod

- Suppose that $\mu(x) = 2ax$, then we (our math friends) know that:

$$F(x) = \int \mu(x)dx = ax^2 + C$$

- We want:

$$\lim_{\Delta x \to 0} \sum_{i=0}^{i=N} \mu(x_i)\Delta x = \int_{x=0}^{x=L} \mu(x)dx$$

- We can thus evaluate:

$$\int_{x=0}^{x=L} \mu(x)dx = F(x=L) - F(x=0) = aL^2 - 0 = aL^2$$

The infinitesimals

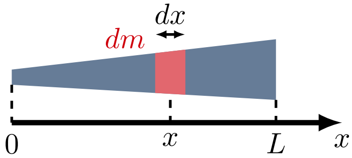
- We can think of an *infinitesimally small* rod of mass, dm, and length, dx:

$$dm = \mu(x)dx$$

- The total mass of the rod, M, is the sum of the masses of the infinitesimal rods:

$$M = \sum dm = \int_0^M dm = \int_0^L \mu(x)dx$$

- With practice, you will get used to just writing everything with infinitesimals rather than a sum and taking the limits!
- This also makes it clear that **an integral is a sum**



A non-uniform rod of mass, M, and length, L, modeled as the sum of many "mini rods" of length Δx and mass Δm_i.

Compare with:

$$\Delta m_i = \mu(x)\Delta x$$

$$M = \sum_{i=0}^{i=N} \Delta m_i = \lim_{\Delta x \to 0} \sum_{i=0}^{i=N-1} \mu(x_i)\Delta x$$